HARD Mock Set 5 — Accenture Technical (Above-Level)

Level: Harder than the 10 uploaded sets. Unique questions — nothing here repeats Sets 1–4.

Format: 50 questions · Time: 50 minutes, closed book · No negative marking.

Section	Area	Qs

A	MS Office & Common Applications	8
B	Networking, Security & Cloud	12
C	CS Core — DSA / DBMS / OOP / HTML-CSS-JS	12
D	Pseudocode & Code Tracing	14
E	Applied — SQL & Code Output	4

Section A — MS Office & Common Applications (8)

Q1. A1:A6 = running, run, jogging, walk, swimming, swim. What does `=COUNTIF(A1:A6, "*ing*")` return?

A) 6

B) 2

C) 3

D) 0

Q2. A1 contains the formula `=IF(B1>10,"",B1)`. What does `=ISBLANK(A1)` return?

A) TRUE, because the cell looks empty

B) TRUE, because the result is an empty string

C) `#N/A`

D) FALSE — a formula (even one returning `""`) means the cell is **not** blank

Q3. A1:A5 = 1, `"abc"`, 3, an empty cell, 5. What does `=COUNT(A1:A5)` return?

A) 5

B) 3

C) 4

D) 2

Q4. You sort a table and the header row ends up in the middle of the sorted data. What did you miss?

A) You must sort by more than one column

B) You must convert the range to a Table first

C) Ticking **"My data has headers"** in the Sort dialog

D) Selecting the whole sheet before sorting

Q5. Which PowerPoint feature records both **timings and narration** so the show can run unattended?

A) Rehearse Timings

B) Presenter View

C) Set Up Slide Show

D) Record Slide Show

Q6. In MS Word, how do you insert a date that **updates automatically** every time the document is printed?

A) Type the date manually

B) Insert → Date & Time → tick **"Update automatically"** (inserts a field)

C) Insert → Header → Page Number

D) Insert → Quick Parts → Document Property

Q7. Which statement about SSDs vs HDDs is correct?

A) An HDD has no moving parts

B) An SSD always has a larger capacity than an HDD

C) An SSD has no moving parts, much faster random access, and a finite number of write cycles

D) SSDs are magnetic and need defragmentation

Q8. In most browsers, which shortcut opens a **private/incognito** window?

A) Ctrl + Shift + N

B) Ctrl + N

C) Ctrl + Shift + P

D) Alt + N

Section B — Networking, Security & Cloud (12)

Q9. Which CIDR block covers **all** private "Class B" addresses (172.16.0.0 – 172.31.255.255)?

A) 172.16.0.0/16

B) 172.16.0.0/20

C) 172.16.0.0/24

D) 172.16.0.0/12

Q10. Which of these is **not** a private (RFC 1918) address?

A) 10.10.10.10

B) 192.168.1.1

C) 172.32.0.1

D) 172.20.5.5

Q11. Which TCP mechanism stops a fast sender from overwhelming a slow receiver?

A) Three-way handshake

B) Sliding-window flow control (the receiver advertises its window size)

C) Slow start only

D) Checksums

Q12. Which IP header field prevents packets from looping forever, and is also what `traceroute` exploits?

A) Protocol

B) Fragment offset

C) DSCP

D) TTL (Time To Live)

Q13. `traceroute` / `tracert` maps a path by relying on which ICMP message?

A) Time Exceeded (TTL expired in transit)

B) Echo Reply

C) Destination Unreachable (port unreachable)

D) Redirect

Q14. Which HTTP version multiplexes many requests over a single connection using **binary frames**?

A) HTTP/1.0

B) HTTP/1.1

C) HTTP/2

D) HTTP/0.9

Q15. What is the main reason to implement VLANs?

A) To increase the size of the collision domain

B) To split one Layer-2 network into smaller broadcast domains (performance + security)

C) To replace routers entirely

D) To assign public IP addresses to hosts

Q16. Which protocol provides **ticket-based mutual authentication** in Windows Active Directory domains?

A) RADIUS

B) LDAP only

C) NTLM

D) Kerberos

Q17. Managing servers through machine-readable definition files stored in version control is called:

A) Infrastructure as Code (IaC)

B) Virtualisation

C) Containerisation

D) Disaster recovery

Q18. Which cloud model bills **per execution** with no servers for you to provision?

A) IaaS

B) PaaS

C) Serverless / FaaS (e.g. AWS Lambda)

D) Colocation

Q19. An attacker registers `bankk-login.com` and emails a link to customers of a real bank. This is:

A) Whaling

B) Phishing with a look-alike (typosquatting) domain

C) Ransomware

D) Pharming only

Q20. Which wireless attack is used to force clients off the network so the attacker can capture the WPA 4-way handshake?

A) Evil twin

B) Bluesnarfing

C) WPS brute force

D) Deauthentication attack

Section C — CS Core: DSA / DBMS / OOP / HTML-CSS-JS (12)

Q21. Which list is in **increasing** order of growth as $n \to \infty$?

A) $\log n$, n, $n \log n$, n^2, 2^n

B) n, $\log n$, $n \log n$, n^2, 2^n

C) $\log n$, n, n^2, $n \log n$, 2^n

D) `log n`, `n log n`, `n`, n^2, 2■

Q22. Shortest path in an **unweighted** graph in O(V + E):

A) Dijkstra with a binary heap

B) Bellman-Ford

C) Floyd-Warshall

D) BFS

Q23. In a **0-indexed array** representation of a binary heap, the children of node `i` are at:

A) `i/2` and `i/2 + 1`

B) `2i` and `2i + 1`

C) `2i + 1` and `2i + 2`

D) `i − 1` and `i + 1`

Q24. Quick sort degrades to $O(n^2)$ when:

A) Every pivot splits the array into 0 and n−1 elements (e.g. already-sorted data with a first/last-element pivot)

B) The array has duplicate keys

C) The array is randomly shuffled

D) The pivot is chosen as the median

Q25. Which SQL statement adds a new column to an existing table?

A) `MODIFY TABLE Emp ADD salary;`

B) `INSERT INTO Emp COLUMN bonus INT;`

C) `UPDATE TABLE Emp ADD COLUMN bonus INT;`

D) `ALTER TABLE Emp ADD COLUMN bonus INT;`

Q26. How do you list customers who have **placed no orders**?

A) `SELECT ... FROM Customers c INNER JOIN Orders o ON c.id = o.cust_id WHERE o.id IS NULL;`

B) `SELECT ... FROM Customers c LEFT JOIN Orders o ON c.id = o.cust_id WHERE o.cust_id IS NULL;`

C) `SELECT ... FROM Customers c RIGHT JOIN Orders o ON c.id = o.cust_id WHERE c.id = NULL;`

D) `SELECT ... FROM Customers WHERE COUNT(orders) = 0;`

Q27. Which index type best supports **range queries** on a sorted column?

A) Bitmap index

B) Hash index

C) B-tree index

D) Full-text index

Q28. A funds-transfer transaction debits one account and credits another; if the credit fails, the debit is rolled back. Which ACID property?

A) Atomicity

B) Consistency

C) Isolation

D) Durability

Q29. Which statement about overloading vs overriding is **FALSE**?

A) Overloading is resolved at compile time from the parameter list

B) Overriding requires the same method name and parameter list

C) Overriding enables runtime polymorphism

D) Overriding is resolved at compile time using the reference type

Q30. In CSS flexbox, which property aligns items along the **main axis**?

A) `align-items`

B) `justify-content`

C) `align-content`

D) `flex-wrap`

Q31. What does `typeof NaN` return in JavaScript?

A) `"number"`

B) `"NaN"`

C) `"undefined"`

D) `"object"`

Q32. Decimal 255 in hexadecimal is:

A) `EF`

B) `F0`

C) `FF`

D) `1FF`

Section D — Pseudocode & Code Tracing (14)

Q33.

```
s ← 0
FOR i ← 1 TO 4
    FOR j ← 1 TO 4
        IF i = j
            s ← s + i
        ELSE IF i < j
            s ← s + 1
        END IF
    END FOR
END FOR
PRINT s
```

A) 10

B) 12

C) 13

D) 16

Q34. (`out` is a **global** string)

```
out ← ""
FUNCTION f(n)
    IF n = 0
        RETURN
    END IF
    out ← out + n
    f(n - 1)
    out ← out + n
END FUNCTION
f(3)
PRINT out
```

A) `123321`

B) 321123

C) 333

D) 123123

Q35.

```
A ← [3, 8, 1, 9, 4]
FOR i ← 0 TO 3
    IF A[i] > A[i + 1]
        swap A[i], A[i + 1]
    END IF
END FOR
PRINT A
```

A) [1, 3, 8, 4, 9]

B) [3, 1, 8, 9, 4]

C) [3, 1, 8, 4, 9]

D) [8, 3, 9, 1, 4]

Q36.

```
n ← 28; c ← 0
FOR i ← 1 TO n
    IF n MOD i = 0
        c ← c + 1
    END IF
END FOR
PRINT c
```

A) 6

B) 5

C) 7

D) 4

Q37.

```
p ← 1
FOR i ← 1 TO 5
    IF i MOD 2 = 1
        p ← p * i
    END IF
END FOR
PRINT p
```

A) 120

B) 8

C) 5

D) 15

Q38.

```
FUNCTION half(a)
    a ← a DIV 2
    RETURN a
END FUNCTION
n ← 37
r ← half(n)
PRINT n, r
```

A) 18 18

B) 37 18

C) 37 37

D) 18 37

Q39.

```
M ← [[1, 3], [5, 7]]
pos_i ← -1; pos_j ← -1
FOR i ← 0 TO 1
    FOR j ← 0 TO 1
        IF M[i][j] = 6
            pos_i ← i
            pos_j ← j
            BREAK
        END IF
    END FOR
END FOR
PRINT pos_i, pos_j
```

A) 0 1

B) 1 0

C) -1 -1

D) 1 1

Q40.

```
i ← 1; s ← 0
WHILE i <= 20
    IF i MOD 3 = 0 AND i MOD 5 <> 0
        s ← s + i
    END IF
    i ← i + 1
END WHILE
PRINT s
```

A) 63

B) 36

C) 45

D) 48

Q41.

```
s ← "Programming"
out ← ""
FOR each ch IN s
    IF ch is a vowel (a, e, i, o, u — case-insensitive)
        out ← out + "*"
    ELSE
        out ← out + ch
    END IF
END FOR
PRINT out
```

A) `Pr*gr*mm*ng`

B) `Pr*gr*mming`

C) `Pr*gr*mm*ng*`

D) `*r*gr*mm*ng`

Q42.

```
FUNCTION cnt(n)
    IF n = 0
        RETURN 0
    END IF
    IF n MOD 2 = 0
        RETURN 1 + cnt(n DIV 10)
    RETURN cnt(n DIV 10)
END FUNCTION
PRINT cnt(24680)
```

A) 4

B) 5

C) 6

D) 0

Q43.

```
A ← [2, 4, 6, 8]
s ← 0
FOR i ← 0 TO 3
    IF A[i] MOD 4 = 0
        s ← s + A[i] * 2
    ELSE
        s ← s + A[i]
    END IF
END FOR
PRINT s
```

A) 20

B) 44

C) 32

D) 28

Q44. Mutual recursion:

```
FUNCTION even(n)
    IF n = 0 RETURN TRUE
    RETURN odd(n - 1)
END FUNCTION
FUNCTION odd(n)
    IF n = 0 RETURN FALSE
    RETURN even(n - 1)
END FUNCTION
PRINT even(7)
```

A) TRUE

B) Compiler error — mutual recursion is not allowed

C) TRUE (7 is odd, so `even(7)` returns TRUE)

D) FALSE

Q45.

```
stack ← empty
FOR each x IN [5, 2, 9, 1]
    IF x MOD 2 = 1
        PUSH x ONTO stack
    ELSE IF stack is not empty
        POP stack
    END IF
END FOR
PRINT SIZE(stack), TOP(stack)
```

A) 2 1

B) 3 1

C) 1 9

D) 2 9

Q46.

```
a ← "listen"
b ← "silent"
count ← 0
FOR each ch IN a
    IF b CONTAINS ch
        count ← count + 1
    END IF
END FOR
PRINT count
```

A) 5

B) 6

C) 4

D) 0

Section E — Applied: SQL & Code Output (4)

Q47. Which query returns the **three highest-paid** employees?

A) `SELECT * FROM Emp ORDER BY salary DESC LIMIT 3, 1;`

B) `SELECT * FROM Emp WHERE salary = MAX(salary) LIMIT 3;`

C) `SELECT * FROM Emp ORDER BY salary DESC LIMIT 3;`

D) `SELECT * FROM Emp ORDER BY salary LIMIT 3;`

Q48. `SELECT COUNT(*), SUM(salary) FROM Emp WHERE dept = 'IT';` when **no** employee is in IT. What is returned?

A) `0` and `NULL`

B) `0` and `0`

C) `NULL` and `NULL`

D) No rows at all

Q49. Java:

```
int count = 0;
for (int i = 0; i < 6; i++) {
    if (i % 2 == 0) continue;
    count += i;
}
System.out.println(count);
```

A) 6

B) 15

C) 12

D) 9

Q50. Which CSS centres a **block-level** element horizontally inside its parent?

A) `text-align: center;` on the element

B) `margin: 0 auto;` together with a defined `width`

C) `align: middle;`

D) `position: center;`

ANSWER KEY

Do not look here until you have finished all 50 questions.

— Answer Key for Set 5

Quick grid

Q	1	2	3	4	5	6	7	8	9	10
Ans	C	D	B	C	D	B	C	A	D	C

Q	11	12	13	14	15	16	17	18	19	20
Ans	B	D	A	C	B	D	A	C	B	D

Q	21	22	23	24	25	26	27	28	29	30
Ans	A	D	C	A	D	B	C	A	D	B

Q	31	32	33	34	35	36	37	38	39	40
Ans	A	C	D	B	C	A	D	B	C	D

Q	41	42	43	44	45	46	47	48	49	50
Ans	A	B	C	D	A	B	C	A	D	B

Explanations

Q1 — C. `*` matches any run of characters, so `*ing*` hits running, jogging, swimming → **3**. (This is a substring test, not "ends with ing".)

Q2 — D. `ISBLANK` is FALSE for any cell containing a formula, even one that returns `""`. Use `=A1=""` or `LEN(A1)=0` to test for an empty-looking result.

Q3 — B. COUNT counts only cells containing **numbers** → 1, 3, 5 → **3**. Use COUNTA for non-empty cells (4 here, ignoring the truly blank cell).

Q4 — C. "My data has headers" excludes the header row from the sort keys so it stays on top.

Q5 — D. Record Slide Show captures narration + timings per slide. Rehearse Timings records timings only.

Q6 — B. With "Update automatically" Word inserts a `DATE`/`PRINTDATE` field that refreshes on print — unlike typed text.

Q7 — C. SSD = flash, no moving parts, very low random-access latency, finite P/E write cycles. HDDs are magnetic, do benefit from defragmentation, and are cheaper per GB.

Q8 — A. Ctrl + Shift + N = new private/incognito window; Ctrl + N = new normal window; Ctrl + Shift + P = new private window in Firefox.

Q9 — D. 172.16.0.0 – 172.31.255.255 = **172.16.0.0/12** (RFC 1918). /16 covers only 172.16.x.x.

Q10 — C. Private ranges: 10/8, 172.16/12, 192.168/16. 172.32.x is **outside** 172.16–172.31 → public.

Q11 — B. The receiver advertises a window size (flow control). Congestion control (slow start, congestion avoidance) is about the *network*, not the receiver.

Q12 — D. TTL is decremented at every hop; at 0 the packet is discarded. That is the basis of traceroute.

Q13 — A. Traceroute sends packets with increasing TTL and collects ICMP **Time Exceeded** messages from each router.

Q14 — C. HTTP/2 adds binary framing, multiplexing, header compression and server push over one connection. HTTP/1.1 serialises requests per connection (head-of-line blocking).

Q15 — B. VLANs create separate Layer-2 broadcast domains on the same physical switches — smaller failure/storm domains and segmentation without extra hardware.

Q16 — D. Kerberos issues TGT/service tickets and authenticates both the user and the service. NTLM is the older challenge-response protocol; RADIUS is for network access.

Q17 — A. IaC (Terraform, ARM, CloudFormation) describes infrastructure in declarative files kept in version control.

Q18 — C. Serverless/FaaS — you deploy functions and pay per invocation and duration, with no server to manage.

Q19 — B. A look-alike registered domain used to harvest credentials = phishing (typosquatting). Pharming poisons name resolution instead.

Q20 — D. A deauth flood disconnects clients; when they reconnect the attacker captures the 4-way handshake for an offline passphrase attack.

Q21 — A. O(1) < O(log n) < O(n) < O(n log n) < O(n^2) < O(n^3) < O(2■) < O(n!).

Q22 — D. BFS gives shortest paths in unweighted graphs in O(V + E). Dijkstra is for weighted graphs (non-negative).

Q23 — C. 0-indexed: left child `2i + 1`, right child `2i + 2`, parent ⌊`(i − 1)/2`⌋.

Q24 — A. Worst case = maximally unbalanced partitions → O(n²) (e.g. sorted input with first/last pivot). Randomised or median-of-three pivots avoid it in practice.

Q25 — D. `ALTER TABLE ... ADD COLUMN` (sometimes just `ADD`) changes a table's structure. INSERT adds rows; UPDATE changes values.

Q26 — B. LEFT JOIN keeps all customers; unmatched rows have NULL for the orders columns, so `WHERE o.cust_id IS NULL` isolates them. `= NULL` never matches — always use `IS NULL`.

Q27 — C. B-tree indexes keep keys sorted, so they support `=`, `<`, `>` and `BETWEEN`/range scans. Hash indexes support equality only.

Q28 — A. Atomicity = all-or-nothing: if any step fails, the whole transaction is rolled back.

Q29 — D. That is the false one: **overriding** is resolved at *runtime* by the object's actual type (dynamic dispatch); overloading is resolved at compile time.

Q30 — B. `justify-content` positions items along the **main axis**; `align-items` works on the **cross axis**.

Q31 — A. `typeof NaN === "number"` — a famous JS quirk. Use `Number.isNaN()` to test for NaN.

Q32 — C. 255 = 15×16 + 15 = **0xFF**.

Q33 — D. Per row i: i = 1 → 1 + 3 = 4; i = 2 → 0 + 2 + 2 = 4; i = 3 → 0 + 0 + 3 + 1 = 4; i = 4 → 4 → total **16**.

Q34 — B. The append happens **before and after** the recursive call: "3" + ("2" + ("1" + "1") + "2") + "3" = **321123**. (It is a palindrome, not 123321!)

Q35 — C. Only adjacent swaps: 3↔8 no, 8↔1 yes → [3,1,8,9,4], 8↔9 no, 9↔4 yes → **[3, 1, 8, 4, 9]** — one pass of bubble sort, which does *not* fully sort the array.

Q36 — A. Divisors of 28: 1, 2, 4, 7, 14, 28 → **6**.

Q37 — D. Odd i only: 1 × 3 × 5 = **15** (note `p` starts at 1, so i = 1 contributes nothing).

Q38 — B. Pass-by-value: `a` is a local copy, so the caller's `n` stays **37**; the return value is 37 DIV 2 = **18**.

Q39 — C. 6 is not in the matrix, so the sentinels are never updated → **-1 -1**. (Note the inner BREAK only exits the inner loop — the outer loop still runs.)

Q40 — D. Multiples of 3 up to 20: 3, 6, 9, 12, 15, 18; drop 15 (also divisible by 5) → 3 + 6 + 9 + 12 + 18 = **48**.

Q41 — A. Vowels o, a, i are replaced → `Pr*gr*mm*ng`. Only lowercase/uppercase vowels count — not `y`.

Q42 — B. All five digits (2, 4, 6, 8, 0) are even; the recursion drops the last digit each time and stops at n = 0 → **5**.

Q43 — C. 2 → +2; 4 → +8; 6 → +6; 8 → +16 → **32**.

Q44 — D. even(7) → odd(6) → even(5) → odd(4) → even(3) → odd(2) → even(1) → odd(0) = **FALSE**. 7 is odd, so `even(7)` is FALSE — the names describe the *remaining* steps.

Q45 — A. Push 5 → [5]; 2 is even and the stack is non-empty → pop → []; push 9 → [9]; push 1 → [9, 1] → size **2**, top **1**.

Q46 — B. Every letter of "listen" (l, i, s, e, t, n) appears in "silent" → **6**. (An anagram test would need counts, not just membership.)

Q47 — C. `ORDER BY salary DESC LIMIT 3`. `LIMIT 3, 1` means "skip 3, take 1" (the 4th row) — a very common trap.

Q48 — A. Aggregate queries without GROUP BY always return exactly one row: `COUNT(*) = 0` and `SUM(...) = NULL` (SUM of an empty set is NULL, not 0).

Q49 — D. `continue` skips even i; odd i are 1, 3, 5 → 1 + 3 + 5 = **9**.

Q50 — B. `margin: 0 auto` plus a fixed width centres a block element. `text-align: center` centres *inline* content (text/images) inside the element.

Score guide (Set 5)

Score	Verdict
43–50	Excellent — safely above Accenture level
34–42	Good — revise the missed areas
25–33	Needs work — revisit the notes, retry after 3 days
< 25	Return to chapters 01–06 and re-sit Sets 1–2